Semester II **B. Tech CSE (AI)**

BCSAI 205: DIGITAL ELECTRONICS

3L+0T+0P+3C **MM 100**

Unit 1: Number Systems And Codes: Introduction to number systems, weighted and non-weighted codes, 1's complement, 2's complement, complement arithmetic
Introduction to Boolean algebra: Postulates and theorems of Boolean algebra, Boolean functions, canonical and standard form, simplification of Boolean function using Boolean laws and theorems

Unit 2:Logic Gates: Diode and transistor as a switch, basic logic gates, derived logic gates, block diagrams and truth tables, logic diagrams from Boolean expression and vice versa, converting logic diagram to universal logic, positive logic, negative logic and mixed logic

Unit 3: Simplification Of Boolean Functions: K-map representation, incompletely specified functions, simplification realization with gates, Quine-McCluskey method
Combinational Logic: Analysis and design of combinational circuits, half adder and full adder, half subtractor and full subtractor, binary serial and parallel adder, BCD adder, binary multipliers, magnitude comparator, decoders, encoders, multiplexers, de-multiplexers

Unit 4:Sequential Circuits: Latches, flip-flops, triggering of the flip-flops, master-slave flip- flop, excitation tables, conversion of the flip-flops, analysis and design of clocked sequential circuits, shift registers, counters

Unit 5: Logic Families: Logic gate characteristics (propagation delay, speed, noise margin, fan-in, fan-out, power dissipation), standard logic families (RTL, DCTL, DTL, TTL, ECL, MOS), tri-state devices
Programmable Logic: Introduction to programmable logic array (PLA) & programmable array logic (PAL

Text/Reference Books:

1. Digital Design, Moris Mano, Pearson Education
2. Digital Fundamental, Floyd and Jain, Pearson Education
3. Digital System: Principles and Applications, Tocci, Pearson Education
4. Digital Electronics, B. P. Singh, DhanpatRai& Sons
5. Modern Digital Electronics, R. P. Jain, Tata McGraw-Hil